FedSynAI

Initial Project Proposal / SRS Document
FedSynAI: A Privacy-Preserving Collaborative AI Framework Using Federated Learning and Generative Synthetic Data for Secure Distributed Machine Learning
Project Duration: 3 Months
Primary Domain: Privacy-Preserving AI
Secondary Domains: Federated Learning, Generative AI, Cybersecurity, Responsible AI
Project Level: Advanced B.E. AIML Major Project
Reviews: Review 0 → Review 1 → Review 2
Research Potential: High

Initial Project Proposal

Organizations such as hospitals, banks, insurance companies, and research institutions possess valuable datasets that cannot be freely shared because of privacy, security, and regulatory constraints. Federated Learning (FL) enables organizations to collaboratively train machine-learning models without transferring their raw data, but model updates may still expose sensitive information and non-IID data distributions can reduce global model performance. Generative synthetic data can support privacy-preserving data sharing and model development, but synthetic datasets must be evaluated for statistical fidelity, privacy leakage, and downstream utility. **FedSynAI** proposes an integrated framework combining Federated Learning with Generative AI-based synthetic data generation to enable secure collaborative machine learning across distributed organizations. The system will simulate multiple data-owning clients, train local models, aggregate model updates, generate synthetic tabular data, and incorporate privacy protection through Differential Privacy and, where feasible, secure aggregation. The framework will evaluate privacy leakage, data heterogeneity, synthetic-data quality, predictive utility, fairness, and communication overhead. Experiments will compare centralized learning, conventional FL, privacy-preserving FL, synthetic-data-based learning, and the proposed FedSynAI framework. The project will be implemented and evaluated within three months through three phased reviews: **Review 0 – problem definition and architecture, Review 1 – working FL and synthetic-data prototype, and Review 2 – integrated system, quantitative evaluation, and final demonstration.**

1. Introduction

The increasing use of Artificial Intelligence in healthcare, finance, insurance, cybersecurity, and other sensitive domains has created a strong requirement for collaborative machine learning without direct sharing of confidential data. Conventional centralized machine learning requires organizations to collect and transfer data to a common location, creating risks associated with privacy, security, regulatory compliance, and unauthorized access.

Federated Learning addresses part of this problem by keeping training data at individual clients and exchanging model information instead of raw records. However, recent research shows that federated learning is not inherently private: model updates can potentially reveal information through membership-inference, reconstruction, poisoning, and other attacks.

Synthetic data provides another approach in which artificial records are generated to preserve useful characteristics of the original data while reducing direct exposure of individual records. However, synthetic data can still suffer from poor fidelity, bias, privacy leakage, and reduced downstream utility. CTAB-GAN+ demonstrates the importance of explicitly studying the trade-off between privacy and utility when generating tabular synthetic data.

FedSynAI therefore combines these approaches into one experimental framework.

1.1 Motivation

The major motivation for this project is the **data-sharing barrier in privacy-sensitive AI applications**.

Organizations may possess complementary datasets, but direct data exchange may be unacceptable or prohibited. Federated Learning allows collaborative model training while keeping data locally, whereas synthetic data can facilitate controlled data generation and experimentation.

However, these approaches introduce important research questions:

- How much privacy is actually achieved?
- Can model updates leak information?
- How does non-IID data affect the global model?
- How realistic are generated synthetic records?
- Can synthetic data provide comparable predictive utility?
- What is the privacy–utility trade-off?
- What security mechanisms can reduce leakage?

Recent research identifies precisely these privacy, security, heterogeneity, and utility challenges in FL.

2. Literature Survey

The literature has been divided into five areas:

1. Federated Learning
2. Privacy-preserving Federated Learning
3. Federated Learning security attacks
4. Synthetic data generation
5. Collaborative/federated synthetic data

2.1 Literature Review – 20 Recent Papers

No.	Year	Paper	Major Contribution	Relevance to FedSynAI
1	**2022**	**Federated Synthetic Data Generation with Differential Privacy**	Introduces private FL-GAN for generating synthetic data while protecting training data through DP.	Direct foundation for FL + synthetic data.
2	**2022**	**An Effective and Practical Gradient Inversion Attack**	Demonstrates that shared gradients can potentially reveal private training information.	Motivates privacy-leakage evaluation.
3	**2023**	**Recent Advances on Federated Learning: A Systematic Survey**	Reviews architectures, algorithms, applications and challenges of FL.	Provides overall FL foundation.
4	**2023**	**Active Membership Inference Attack under Local Differential Privacy in Federated Learning**	Demonstrates that even FL with local DP can remain vulnerable to active membership inference.	Supports MIA/privacy testing.

5	**2023**	**On the Effectiveness of Partial Variance Reduction in Federated Learning with Heterogeneous Data**	Addresses model drift caused by heterogeneous client data.	Supports non-IID experiments.
6	**2023**	**The Aggregation–Heterogeneity Trade-off in Federated Learning**	Studies the relationship between aggregation and data heterogeneity.	Supports heterogeneity analysis.
7	**2023**	**CTAB-GAN+: Enhancing Tabular Data Synthesis**	Improves tabular synthetic data generation and incorporates DP-SGD.	Main synthetic-data baseline.
8	**2023**	**MIA-Leak: Exploring Membership Inference Attacks in Federated Learning Systems**	Investigates membership inference using generative approaches.	Supports privacy-attack experiments.
9	**2023**	**Perfectly Accurate Membership Inference by a Dishonest Central Server in Federated Learning**	Demonstrates privacy risk from a dishonest FL server.	Motivates server-side privacy protection.
10	**2023**	**Comparative Analysis of Membership Inference Attacks in Federated and Centralized Learning**	Compares privacy vulnerabilities in centralized and federated settings.	Useful for baseline comparison.
11	**2024**	**A Multifaceted Survey on Privacy Preservation of Federated Learning**	Reviews privacy attacks, secret sharing and privacy-preserving FL mechanisms.	Supports privacy architecture.
12	**2024**	**Federated Learning Privacy: Attacks, Defenses, Applications, and Policy Landscape**	Provides comprehensive analysis of FL privacy attacks and defenses.	Defines security/evaluation framework.
13	**2024**	**CaPS: Collaborative and Private Synthetic Data Generation from Distributed Sources**	Generates synthetic data from distributed data holders using MPC and DP.	**Closest research reference to FedSynAI.**
14	**2024**	**Privacy Mechanisms and Evaluation Metrics for Synthetic Data Generation: A Systematic Review**	Reviews privacy mechanisms and evaluation metrics for synthetic data.	Supports quality/privacy metrics.
15	**2024**	**Fairness and Privacy Preserving in Federated Learning: A Survey**	Studies privacy and fairness together in FL.	Supports responsible-AI evaluation.
16	**2024**	**A Comprehensive Review on Federated Learning for Data-Sensitive Applications**	Reviews FL for healthcare, finance and other sensitive applications.	Supports application justification.
17	**2025**	**The Federation Strikes Back: A Survey of Federated Learning Privacy Attacks, Defenses,**	Reviews privacy attacks, defenses and regulatory issues.	Strengthens privacy/security component.

		Applications, and Policy Landscape		
18	**2025**	**Backdoor Attacks and Defense Mechanisms in Federated Learning: A Survey**	Reviews backdoor threats and defenses in FL.	Supports optional security extension.
19	**2026**	**On the Security and Privacy of Federated Learning: A Survey with Attacks, Defenses, Frameworks, Applications, and Future Directions**	Reviews 203 papers covering attacks, defenses, privacy, non-IID and security.	Latest comprehensive security foundation.
20	**2026**	**A Survey of Federated Learning: Advances in Architecture, Synchronization, and Security Threats**	Reviews FL architecture, synchronization, data distribution and security.	Supports final architecture and research gap.

Literature synthesis

The literature establishes four important findings.

First, Federated Learning reduces the need to transfer raw data but does not guarantee complete privacy. Model updates can be attacked through membership inference, gradient inversion, poisoning and other techniques.

Second, non-IID data is a fundamental FL challenge. Different organizations may have different sample sizes, feature distributions, class distributions and data quality. This can cause local-model drift and degrade global performance.

Third, synthetic data can support privacy-preserving data sharing, but realistic-looking data is not automatically private. Recent work emphasizes the need to measure privacy, statistical similarity and downstream ML utility together.

Fourth, CaPS provides particularly strong evidence for the need for collaborative private synthetic-data generation from distributed data holders. It combines secure multi-party computation for input privacy with differential privacy for output privacy.

2.2 Limitations of Existing Systems

Based on the reviewed literature, the following limitations are relevant.

1. Federated Learning is not completely private

Although raw data remains local, gradients or model parameters can leak information.

2. Synthetic data can leak information

A generative model can potentially memorize or reproduce characteristics of training records.

3. Privacy–utility trade-off

Adding stronger differential privacy can reduce model or synthetic-data utility. CTAB-GAN+ explicitly evaluates this trade-off.

4. Non-IID data

Different clients may have significantly different distributions, making global model convergence difficult.

5. Security threats

FL systems may be vulnerable to:

- Membership inference
- Gradient inversion
- Data poisoning
- Backdoor attacks

- Malicious clients
- Malicious server

Recent 2026 research continues to identify these as major FL challenges.

6. Fragmented evaluation

Many studies concentrate on either:

- privacy,
- synthetic data quality,
- FL accuracy, or
- security,

rather than providing a unified experimental framework covering all these dimensions.

2.3 Outcome of Literature Survey

The literature suggests a need for an integrated framework that evaluates:

Federated Learning + Synthetic Data + Privacy + Security + Data Heterogeneity + ML Utility

FedSynAI addresses this gap by providing a common experimental environment where distributed clients can collaboratively train models and generate synthetic data without directly exchanging raw records.

The project will particularly extend the ideas demonstrated by private FL-GAN and CaPS by creating a student-implementable framework combining **federated model training, synthetic data generation, differential privacy and quantitative privacy/utility evaluation**.

3. Problem Statement

Develop FedSynAI, a privacy-preserving collaborative AI framework that combines Federated Learning and Generative AI-based synthetic data generation to enable multiple organizations to collaboratively develop and evaluate machine-learning models without directly sharing sensitive raw data. The framework shall address non-IID data heterogeneity, privacy leakage, synthetic-data quality, predictive utility and security through appropriate privacy-preserving mechanisms and quantitative evaluation.

3.1 Objectives

Your format says **maximum 2 objectives**, so although the broader project has four technical goals, I recommend consolidating them into **two measurable objectives**.

Objective 1

To develop a federated collaborative learning and synthetic-data generation framework that enables distributed clients to train machine-learning models and generate useful synthetic tabular data without sharing their raw datasets.

Objective 2

To evaluate and improve the proposed framework with respect to privacy leakage, non-IID data heterogeneity, synthetic-data quality, predictive utility and security using quantitative privacy and machine-learning metrics.

These two objectives are broad enough to cover all four technical dimensions while complying with the **maximum-two-objective** project format.

4. Proposed Methodology

4.1 Overall Method

The project will use the following sequence:

Dataset Selection

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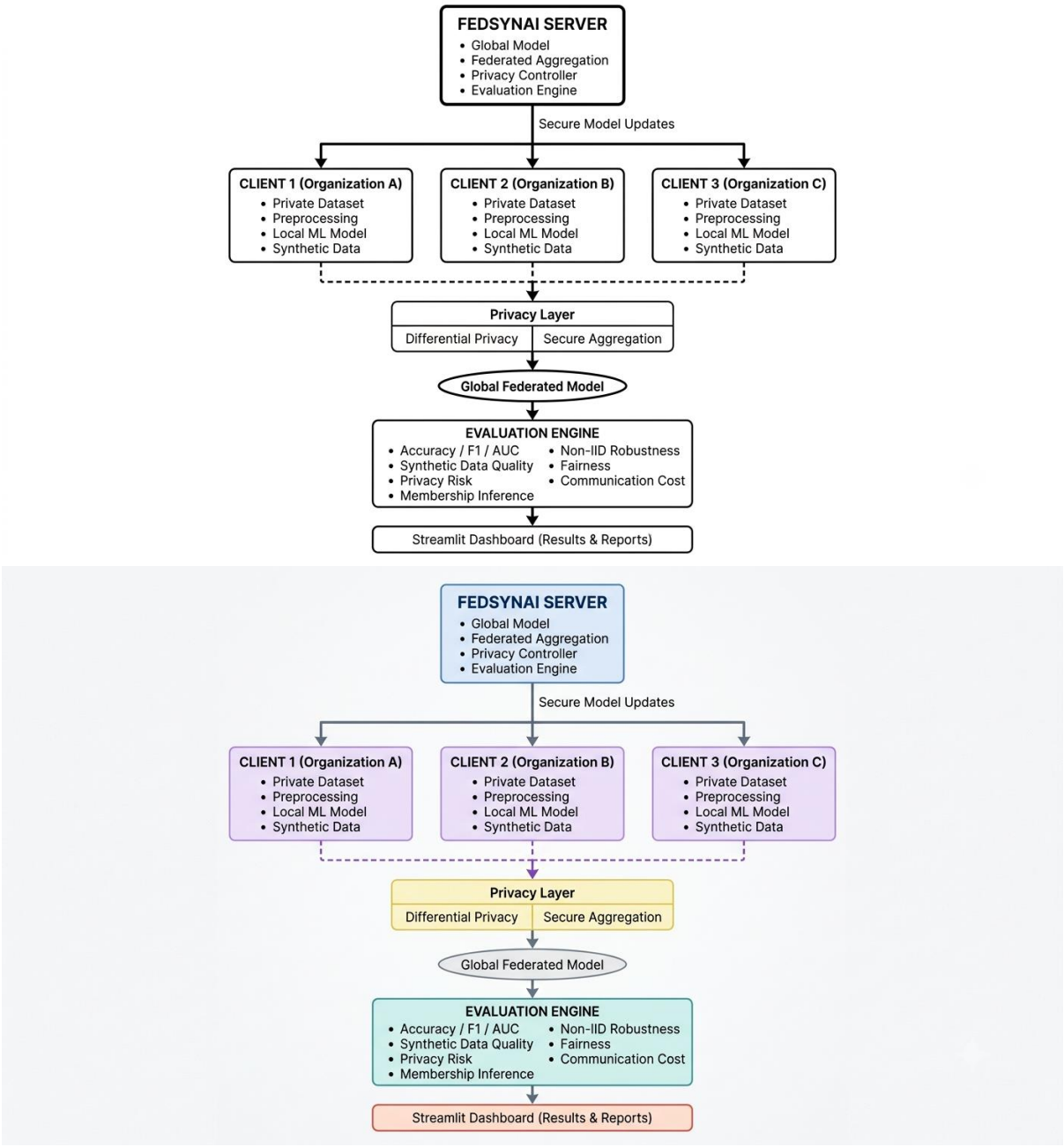
Data Preprocessing

↓

Client/Organization Simulation

↓
IID + Non-IID Data Distribution
↓
Centralized ML Baseline
↓
Federated Learning Baseline
↓
Synthetic Data Generation
↓
Differential Privacy
↓
Secure Model Aggregation
↓
Privacy Attack Evaluation
↓
Synthetic Data Quality Evaluation
↓
Predictive Utility Evaluation
↓
Comparison Dashboard

4.2 Architecture Diagram



4.3 Experimental Comparison

The students should implement at least **five configurations**:

Experiment	Description
E1	Centralized ML using combined real data
E2	Standard Federated Learning
E3	Federated Learning with non-IID clients
E4	Privacy-preserving FL with Differential Privacy
E5	**FedSynAI – FL + Synthetic Data + Privacy Protection**

This makes the final project experimentally meaningful.

4.4 Synthetic Data Module

Recommended implementation:

Primary model
CTAB-GAN+
because it supports:

- Continuous variables
- Categorical variables
- Imbalanced distributions
- Classification
- Regression
- Differentially private training

CTAB-GAN+ explicitly evaluates both statistical similarity and downstream ML utility under different privacy budgets.
Optional comparison
CTGAN vs CTAB-GAN+
This can be added if time permits.

4.5 Federated Learning Module
Baseline
FedAvg
Optional advanced method
FedProx
for non-IID data.
Students should compare:
FedAvg
vs
FedProx
vs
FedSynAI
The comparison should focus on:

- Accuracy
- F1
- convergence
- client variability
- communication rounds

4.6 Privacy Module
Primary
Differential Privacy
Measure:

- Privacy budget ε
- Noise multiplier
- Accuracy
- F1
- Privacy leakage

Optional
Secure Aggregation
If implementation complexity becomes high, secure aggregation can be implemented as an extension/simulation rather than the core contribution.

4.7 Security Evaluation

The project should include at least **one privacy attack**.

Recommended:

Membership Inference Attack

Students compare:

Without DP → Attack Success

With DP → Attack Success

FedSynAI → Attack Success

Recent studies demonstrate that FL models can remain susceptible to membership inference even when privacy mechanisms are applied, making this a meaningful evaluation component.

4.8 Evaluation Metrics

Machine Learning

Metric	Purpose
Accuracy	Overall prediction
Precision	Positive prediction quality
Recall	Detection capability
F1-score	Balanced classification
ROC-AUC	Class discrimination

Synthetic Data

Metric	Purpose
Distribution similarity	Compare real vs synthetic
Correlation similarity	Preserve feature relationships
KS statistic	Distribution difference
Wasserstein distance	Distribution similarity
TSTR	Train Synthetic, Test Real
TRTS	Train Real, Test Synthetic

Privacy

Metric	Purpose
ε	Differential privacy guarantee
Membership inference accuracy	Privacy leakage
Attack success rate	Security assessment
Re-identification risk	Record-level privacy

FL

Metric	Purpose
Communication rounds	Efficiency
Convergence	Training behavior
Client accuracy	Client-level performance
Global accuracy	Overall performance
Communication overhead	Scalability

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5. System Requirement Specifications

5.1 Functional Requirements

FR1 – Dataset Management

The system shall allow administrators/researchers to:

- Upload/select datasets.
- Preprocess data.
- Handle missing values.
- Normalize numerical attributes.

- Encode categorical attributes.

FR2 – Client Simulation

The system shall simulate multiple organizations/clients.

Example:

Client 1 – Hospital A

Client 2 – Hospital B

Client 3 – Hospital C

Client 4 – Hospital D

Raw data shall remain within the simulated client environment.

FR3 – Data Distribution

The system shall support:

- IID distribution
- Non-IID distribution
- Unequal client sizes
- Class imbalance

FR4 – Centralized Baseline

The system shall provide a centralized ML model for comparison.

FR5 – Federated Learning

The system shall:

- Initialize a global model.
- Send the model to clients.
- Train locally.
- Receive model updates.
- Aggregate updates.
- Update the global model.

FR6 – Synthetic Data Generation

The system shall generate synthetic tabular data using CTAB-GAN+ or another selected generator.

FR7 – Privacy Protection

The system shall support Differential Privacy.

The system should record the selected privacy parameters and corresponding model performance.

FR8 – Security Evaluation

The system shall support at least one privacy attack/evaluation mechanism such as membership inference.

FR9 – Evaluation

The system shall calculate:

- Predictive performance
- Privacy risk
- Synthetic-data quality
- Communication cost
- Client/global performance

FR10 – Visualization

The system shall provide a dashboard displaying:

- Accuracy vs rounds
- F1 vs rounds
- Real vs synthetic distributions
- Privacy budget vs accuracy
- Attack success rate
- Client performance
- Communication overhead

FR11 – Report Generation
The system shall generate a consolidated experimental report.

5.2 Non-Functional Requirements
NFR1 – Privacy
Raw client data shall not be transferred to the central server.
NFR2 – Security
Model updates and communication channels should be protected against unauthorized access.
NFR3 – Performance
The system should complete experiments within reasonable computational time on available laboratory hardware.
NFR4 – Scalability
The architecture should support increasing the number of simulated clients.
NFR5 – Reliability
The system should handle client failures or unavailable clients without complete system failure.
NFR6 – Usability
The dashboard should allow students/researchers to configure experiments without modifying the core source code.
NFR7 – Reproducibility
Experiments should use fixed random seeds, version-controlled code, documented datasets and configuration files.
NFR8 – Maintainability
The system should use modular components for:

- FL
- Synthetic generation
- Privacy
- Security
- Evaluation
- Visualization.

6. Software Requirements

Component	Recommended Requirement
OS	Windows/Linux
Language	Python 3.10+
ML	Scikit-learn / PyTorch
FL	Flower
Synthetic Data	SDV / CTGAN / CTAB-GAN+
Privacy	Opacus / DP implementation
Data Processing	Pandas / NumPy
Visualization	Matplotlib / Plotly
Dashboard	Streamlit
Experiment Tracking	MLflow
Version Control	Git/GitHub
Containerization	Docker – optional
IDE	VS Code / PyCharm / Jupyter

7. Hardware Requirements

Minimum

- Intel i5 / Ryzen 5
- 16 GB RAM
- 512 GB SSD
- Stable internet connection

Recommended

- Intel i7 / Ryzen 7
- 32 GB RAM
- NVIDIA GPU with 6–12 GB VRAM
- 1 TB SSD

For the proposed **tabular-data project**, a GPU is useful for GAN training but is not mandatory for all experiments.

8. Dataset Requirement

For a three-month project, **one primary dataset should be selected**.

Recommended options

Healthcare

- MIMIC-IV
- UCI healthcare datasets

Finance

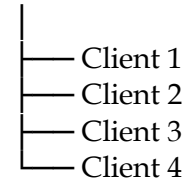
- Credit-risk dataset
- Fraud-detection dataset

General-purpose

- Adult
- Insurance
- Credit
- Loan

The data should be partitioned into simulated organizations:

Dataset



Both IID and non-IID configurations should be evaluated.

9. Three-Month Implementation Timeline

Phase 0 – Proposal and Foundation

Weeks 1–3

Week	Activity	Expected Output
Week 1	Problem identification + literature survey	20-paper literature matrix
Week 2	Dataset + architecture + methodology	Architecture and SRS
Week 3	Centralized ML baseline	Initial baseline results
End Week 3 Review 0		Proposal approval

Review 0

Students must demonstrate:

- Problem statement
- Motivation
- Literature survey

- Research gap
- Objectives
- Dataset
- Architecture
- SRS
- Timeline
- Initial baseline

Phase I – Core Implementation
Weeks 4–7

Week	Activity	Expected Output
Week 4	Data preprocessing + client creation	Multi-client dataset
Week 5	FedAvg implementation	Working FL
Week 6	IID/non-IID experiments	Heterogeneity results
Week 7	CTGAN/CTAB-GAN+	Synthetic-data prototype
End Week 7 Review 1		Working prototype

Review 1
Minimum demonstration:
Centralized ML → Federated Learning → Non-IID FL → Synthetic Data
Students should show preliminary graphs and results.

Phase II – Privacy, Security and Final Evaluation
Weeks 8–12

Week	Activity	Expected Output
Week 8	Differential Privacy	DP-FL
Week 9	Privacy attack	MIA/privacy evaluation
Week 10	Synthetic-data quality + utility	Evaluation metrics
Week 11	Complete FedSynAI comparison	Final experimental results
Week 12	Dashboard + report + paper	Final system
End Week 12 Review 2		Final demonstration

10. Review Milestones

Review	Time	Main Evaluation	Expected Status
Review 0	End of Week 3	Proposal + Literature + SRS	**Problem and methodology approved**
Review 1	End of Week 7	Core implementation	**Working FL + synthetic-data prototype**
Review 2	End of Week 12	Complete system + results	**Integrated FedSynAI + evaluation + report**

11. Expected Final Outcomes
At the end of three months, the students should deliver:
Software

- FedSynAI prototype
- Federated server
- Multiple simulated clients
- Synthetic-data generator

- Privacy module
- Security evaluation module
- Evaluation engine
- Streamlit dashboard

Research

- Literature survey
- Experimental comparison
- Privacy–utility analysis
- Non-IID analysis
- Synthetic-data quality analysis
- Security/privacy attack analysis

Documentation

- SRS
- Project report
- User manual
- Experimental results
- GitHub repository
- Research paper draft

12. Expected Research Contribution

The project should **not claim to invent a new Federated Learning algorithm** unless the students actually develop and experimentally validate one.

The defensible B.E.-level contribution is:

An integrated and experimentally evaluated privacy-preserving framework combining federated learning, synthetic data generation, differential privacy and privacy-risk assessment, with a systematic comparison of predictive utility, synthetic-data quality, privacy leakage and non-IID robustness.

The strongest literature connection is **CaPS**, which demonstrates collaborative and private synthetic-data generation from distributed sources using secure multi-party computation and differential privacy. FedSynAI can use that work as a conceptual reference while implementing a more accessible B.E.-level framework based on federated learning, generative synthetic data and quantitative evaluation.

References — IEEE Style

Below is a **20-paper reference list from 2022–2026**, formatted for inclusion in the proposal.

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